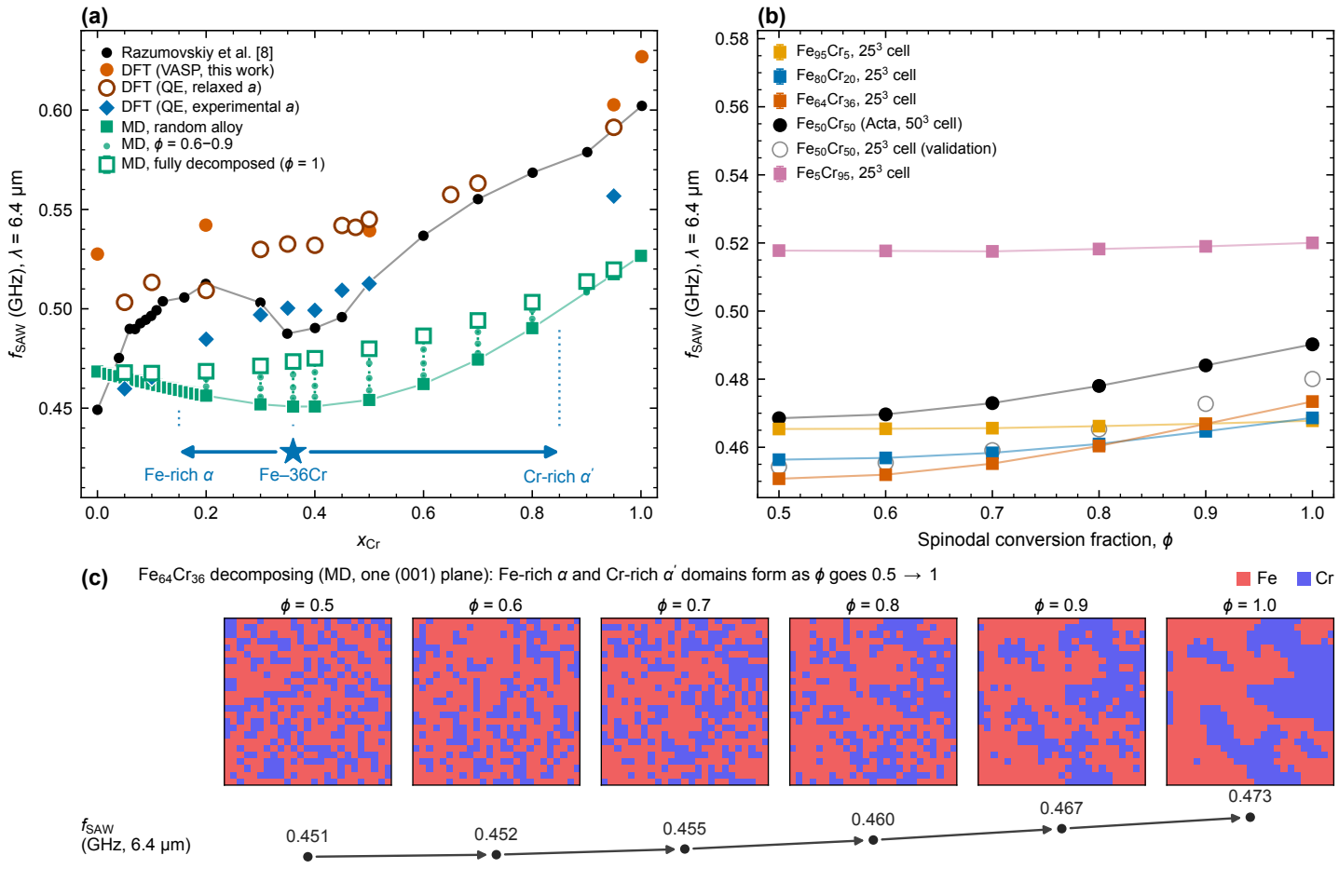

(a)
Razumovskiy et al. [8]
DFT (VASP, this work)
DFT (QE, relaxed a)
DFT (QE, experimental a)
MD, random alloy
MD, ϕ = 0.6–0.9
MD, fully decomposed (ϕ = 1)
Fe-rich α
Fe–36Cr
Cr-rich α′
f_SAW (GHz), λ = 6.4 μm
x_Cr
(b)
Fe95Cr5, 25^3 cell
Fe80Cr20, 25^3 cell
Fe64Cr36, 25^3 cell
Fe50Cr50 (Acta, 50^3 cell)
Fe50Cr50, 25^3 cell (validation)
Fe5Cr95, 25^3 cell
f_SAW (GHz), λ = 6.4 μm
Spinodal conversion fraction, ϕ
(c)
Fe64Cr36 decomposing (MD, one (001) plane): Fe-rich α and Cr-rich α′ domains form as ϕ goes 0.5 → 1
Fe
Cr
ϕ = 0.5
ϕ = 0.6
ϕ = 0.7
ϕ = 0.8
ϕ = 0.9
ϕ = 1.0
f_SAW (GHz, 6.4 μm)
0.451
0.452
0.455
0.460
0.467
0.473